

EXPERIMENT NO :- 2

Title:- Study on pulse code Modulation and Demodulation Technique.

Objective :-

- To study the waveform
- To plot the o/p across each block on the graph paper.

List of Instrument:-

SI No	Name of instrument	Specification	Model No	Manufacturer	Quantity
1	PCM Transmitter & receiver kit	—	572103	scrontech	1
2.	CRO	28MHz dual beam	DSO-5200E	METRAVI	1

Theory:-

pulse code Modulation consists of three steps.

The steps are SAMPLING, QUANTIZATION and ENCODING.

SAMPLING:

The analog signal is sampled according to the Nyquist criteria. The Nyquist criteria states that for faithful reproduction of the band limited signal, the sampling rate must be at least twice the highest frequency component present in the signal. For audio signals the highest frequency component

is 3.4 kHz.

So, Sampling frequency $> 2f_m$

Practically, the sampling frequency is kept slightly more than the required rate. In telephony the standard sampling rate is 8 kHz. Sampling quantifies the instantaneous value of the analog signal point at sampling point to obtain pulse amplitude output.

ALLOCATION OF BINARY CODES:

Each binary word defines a particular narrow range of amplitude level. The sampled value is then approximated to the nearest amplitude level. The sample is then assigned a code corresponding to the amplitude level, which is then transmitted.

QUANTIZATION NOISE:-

As we have seen the signal is approximated to the nearest level (step). Since the level are discrete where as the signal value & its approximated one is random & unpredictable. This is a sort of unwanted, unpredictable, random signal which accompanies the information signal, is termed as Quantization noise.

Quantization noise can be reduced by increasing the number of levels, hence reducing the approximation. But it never to be eliminated. Increasing the no. of levels to reduce without price. Increasing the number bits to represent as a sample increase the system bandwidth requirement.

Experimental Result

RESULT:

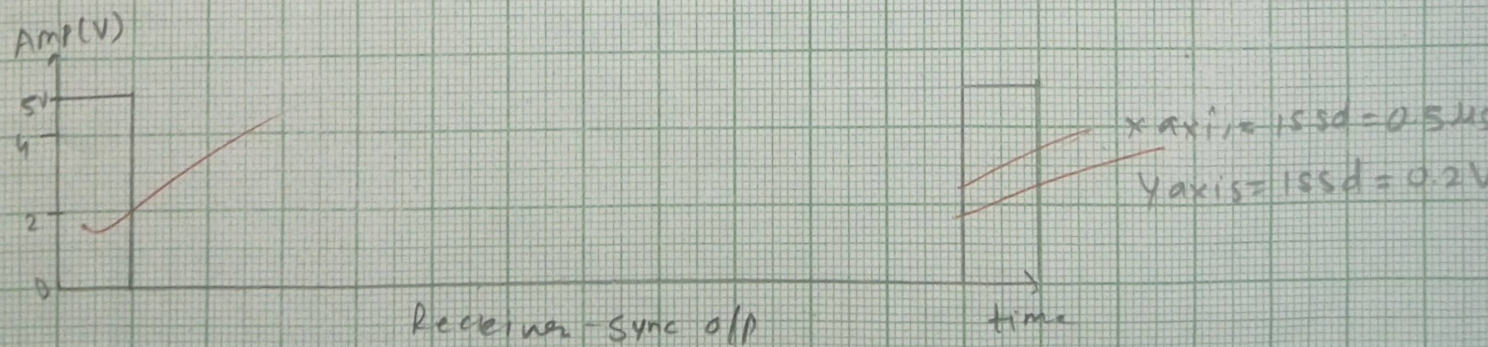
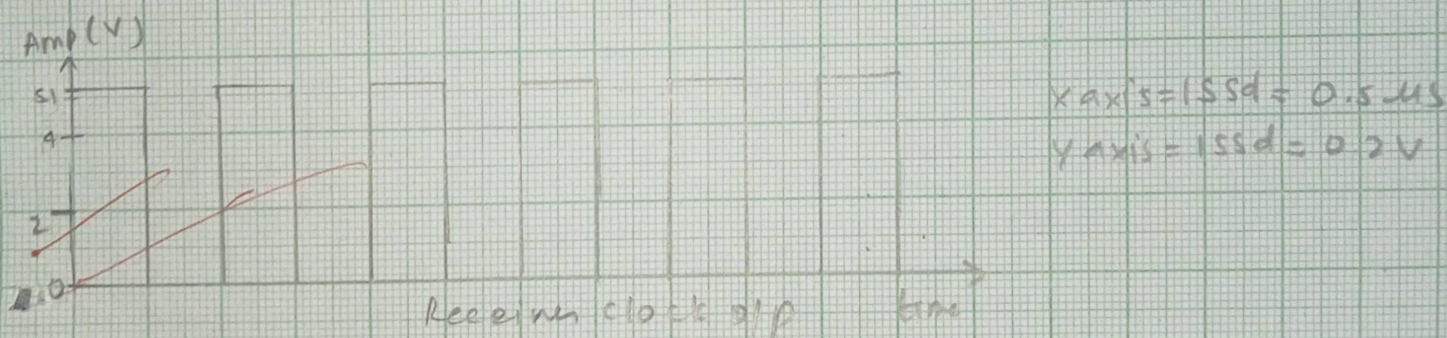
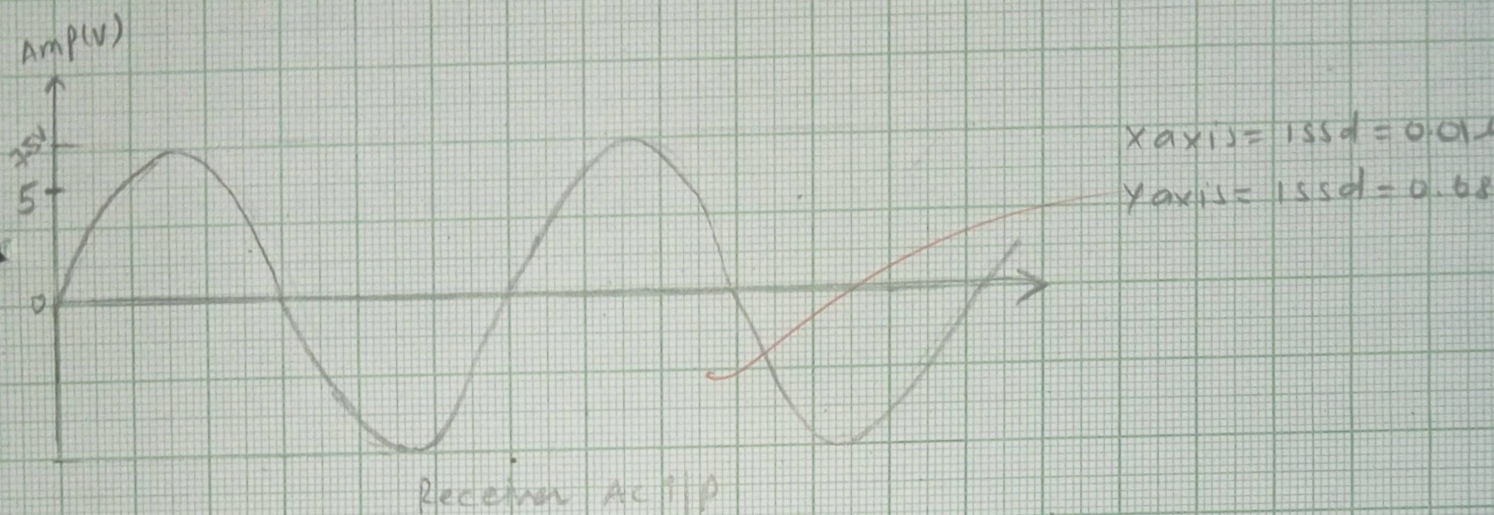
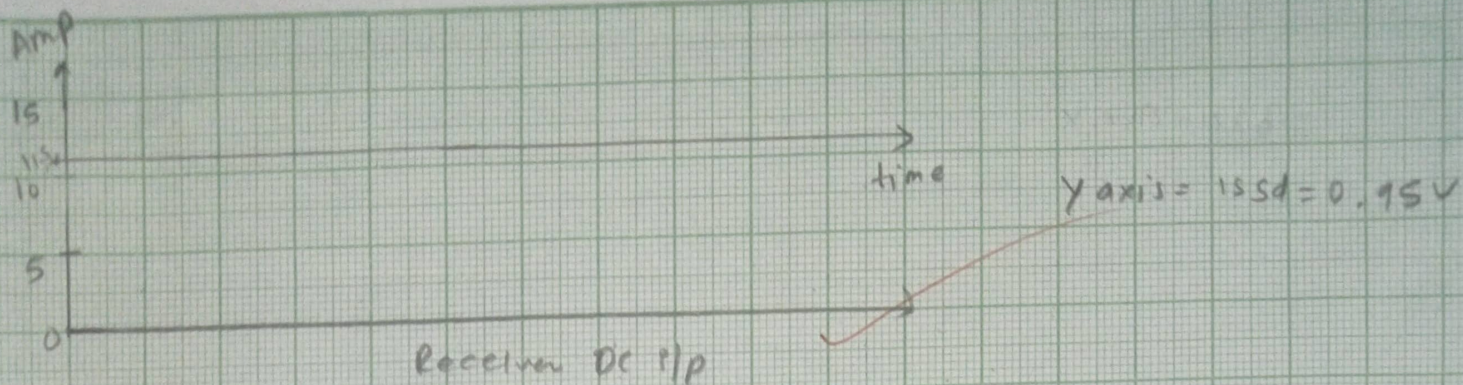
Sl. No.	Name of the blocks	Amplitude	Time	Frequency
1	DC i/p	3.2V	—	—
2	Ac i/p	320mV	14ms	1KHz
3	Tx clock	5.2V	4μs	250KHz
4	Tx sine	5V	64μs	15.6KHz

Sl No	Name of block	Amplitude	Time	frequency
5	PCM	5V	—	—
6	Rx clock	5.2V	4μs	250KHz
7	Rx sync	5V	64μs	15.6KHz
8	DC o/p	23V	—	—
9	Ac o/p	15V	1000μs	1KHz

PCM encoded data = [1011101011010100]

Discussion:- We conclude that the transmitter side sampling for 500KHz and 1KHz signals is done by using 16KHz sampling clock, hereby satisfying the Nyquist criterion. Similarly the multiplexed output observed in the oscilloscope shows the proper alignment of samples in their respective time slots.

At the receiver side the 4 order low pass Butterworth filter is used as reconstruction unit which reproduce the signal (same as that of the transmitter side). It is observed in this case that the reconstructed sine wave has good linearity.



Amplitude = (V)

